

MIRABELLE FENG

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EDUCATION

Carnegie Mellon University

B.S. Computer Science and Arts (BCSA) – GPA 4.0/4.0

Pittsburgh, PA

Aug 2025 – May 2029

- **Coursework:** 15-362 Computer Graphics, 15-210 Parallel & Sequential Data Structures and Algorithms, 15-251 Great Theoretical Ideas in CS, 15-150 Functional Programming, 15-122 Imperative Computation, 21-241 Matrices & Linear Transformations, 60-120 Introduction to Digital Media, 60-212 Creative Coding / Interactivity and Computation

EXPERIENCE

Teaching Assistant, 15-122 Principles of Imperative Computation

Carnegie Mellon University (Spring, Summer, Fall 2026)

Jan 2026 – Present

Pittsburgh, PA

- Lead two weekly recitations and hold office hours for a course serving 1000+ students annually, teaching C programming, data structures, algorithms, and systems concepts
- Since Summer 2026, serve as **grading group lead**, coordinating TAs during staff grading sessions; study rubrics in advance, resolve grading questions, and calibrate scoring to the concepts assessed
- Grade exams and assignments, debug student C code during office hours, and playtest quizzes and homeworks to identify correctness and clarity issues before release

Programmer, Game Creation Society

Carnegie Mellon University

Fall 2026 – Present

Pittsburgh, PA

- Develop core gameplay mechanics and game loops using C# and Unity for Fruit and Flies, leading to smooth player interaction and responsive controls.
- Partner with game designers, artists, and musicians in an agile team environment to translate high-level design vision into functional code
- Manage project source code using Git and GitHub, establishing branching strategies (Git Flow), resolving merge conflicts, and conducting code reviews to maintain repository stability.

Software Developer, ScottyLabs

Carnegie Mellon University

Aug 2025 – May 2026

Pittsburgh, PA

- Built modular, reusable React and Tailwind CSS components for a campus social app prototype, supporting rapid feature iteration across the team
- Implemented responsive UI from Figma files in collaboration with the design team

PROJECTS

Scotty3D Graphics Engine | C++, 15-362 Computer Graphics

Aug 2026 – Present

- Built a full software rasterization pipeline in C++: scene-graph transforms, diamond-exit line drawing, and bounding-box triangle coverage with top-left tie-breaking, plus depth testing and alpha blending
- Implemented perspective-correct barycentric interpolation with UV derivatives, trilinear mip-mapping (generation, sampling, and LOD selection), and configurable supersampling with a custom sample pattern
- Implementing functionality in geometry meshes, including adding, deleting, and transforming vertices, edges and faces

file fisher | Unity, C#, pixel art | blueherr1ng.itch.io/file-fisher

Aug 2026

- Published a 2D pixel-art desktop idler in which a cat pulls random files out of the user's Downloads folder on a user-set interval, as passive file management for people who find organizing files daunting
- Built the basket UI (open, remove, or send a caught file to the trash) and a settings panel controlling fishing frequency; shipped a macOS build to itch.io
- Solo project: wrote all C# game logic and drew all pixel art assets

fish.bowl | TouchDesigner, MediaPipe, real-time 3D

Apr 2026

- Built a hand-tracked interactive installation: a clenched fist spawns a fish into a 3D bowl, and an extended index finger drives a directional "wind" force that pushes fish through the water and can launch them out of the bowl
- Authored the full node network, including 3D scene, models, forces, and gravity restore behavior, on top of the MediaPipe hand-landmark library

TECHNICAL SKILLS

Languages: C, C++, C#, Python, SML, JavaScript, SQL

Graphics & Creative: p5.js, WebGL, Unity, Blender, TouchDesigner, MediaPipe, Photoshop, Illustrator, InDesign, Premiere

Tools: Git/GitHub, VS Code, Figma, Linux/Unix Shell

Activities: Game Creation Society (programmer, *Fruit and Flies*); Computer Graphics Club